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A user-friendly decision support system for supporting grape downy mildew control in Chinese vineyards

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ABSTRACT

Reducing fungicide use is a global priority for agricultural nations. Decision support systems (DSSs) help farmers optimize spraying schedules by assessing crop disease risk. However, a practical DSS specifically designed for grape downy mildew **(*Plasmopara viticola*)** has not yet been reported for China. . In this study, we developed a user-friendly DDS to support grape downy mildew management, which serves as a signal for the first spraying application. Furthermore, the tool employs a binary risk classification system (low vs. high) to recommend optimal spraying timings during the secondary infection period. The DSS has been embedded in the WeChat mini-program "Grape Defender" and deployed across 13 provinces in China. Model evaluation was conducted using two datasets: (a) a multi-year (2021–2024) experimental trial conducted across two cities, and (b) climate data collected from 13 provinces. We then compared a model-based fungicide application approach with a regular method in terms of disease incidence and number of spray applications in the two cities, and then we provide prediction services in 13 major grape-growing provinces across China. The DDS reduced fungicide applications by 43.75% in two experimental cities while maintaining comparable disease control, with increase in disease incidence no exceeding 11%relative to the regular management group. Simulations further projected 73 high-risk infection events across 13 provinces in 2025, suggesting that disease management efforts should primarily focus on June to August rather than May. The **"Grape Defender"** DSS, which will be freely accessible, supports customized strategies tailored to local climates and grape varieties in China.

1. Introduction

Plant disease management plays a critical role in ensuring stable crop production. Grapevine downy mildew, caused by *Plasmopara viticola*, is one of the most economically important diseases in grape-growing regions worldwide, causing significant yield and quality losses when environmental conditions favor disease development [1,2]. Decision support systems (DSSs) have been developed for grape downy mildew in several countries, including Italy [3–5], Spain [6], and India [7], while a commercialized decision-making system has been developed by Pessl Instruments in Austria (Pessl [8]). DSSs such as **GrapeVision** (Bayer) provide user-friendly interfaces for desktop and offline mobile applications in Italy (https://agrocloud.bayer.com/se/en/), and the **GrapeBreed4IPM** project has implemented four DSSs—VitiMeteo [6], Decitrait [9], RIMpro [10], and vite.net [11]—with field trials conducted across four European countries (https://eu-cap-network.ec.europa.eu/projects/). These systems integrate economic costs and climatic factors in disease models, allowing users to determine optimal spraying times to maintain low disease incidence, thereby achieving effective plant disease management [12,13]. Several studies have demonstrated that such simulators reduce fungicide use compared to calendar-based strategies without increasing risks for farmers [14,15].

However, despite these demonstrated advantages, DSSs have not been widely adopted by farm managers and are rarely incorporated into routine management decisions or integrated pest management (IPM) programs [15]. In many cases, these systems primarily provide data for governmental agencies and are validated at specific experimental sites. In Europe, the implementation of IPM principles is the responsibility of regional governments; within each region (NUTS-2 level), plant protection services support farmers through weekly phytosanitary bulletins, with pathogen infection risks reported at the provincial level (NUTS-3) [4]. In China, the National Agro-Tech Extension and Service Center publish the likelihood of major crop disease and pest occurrence, while the Ministry of Agriculture and Rural Affairs and local plant protection

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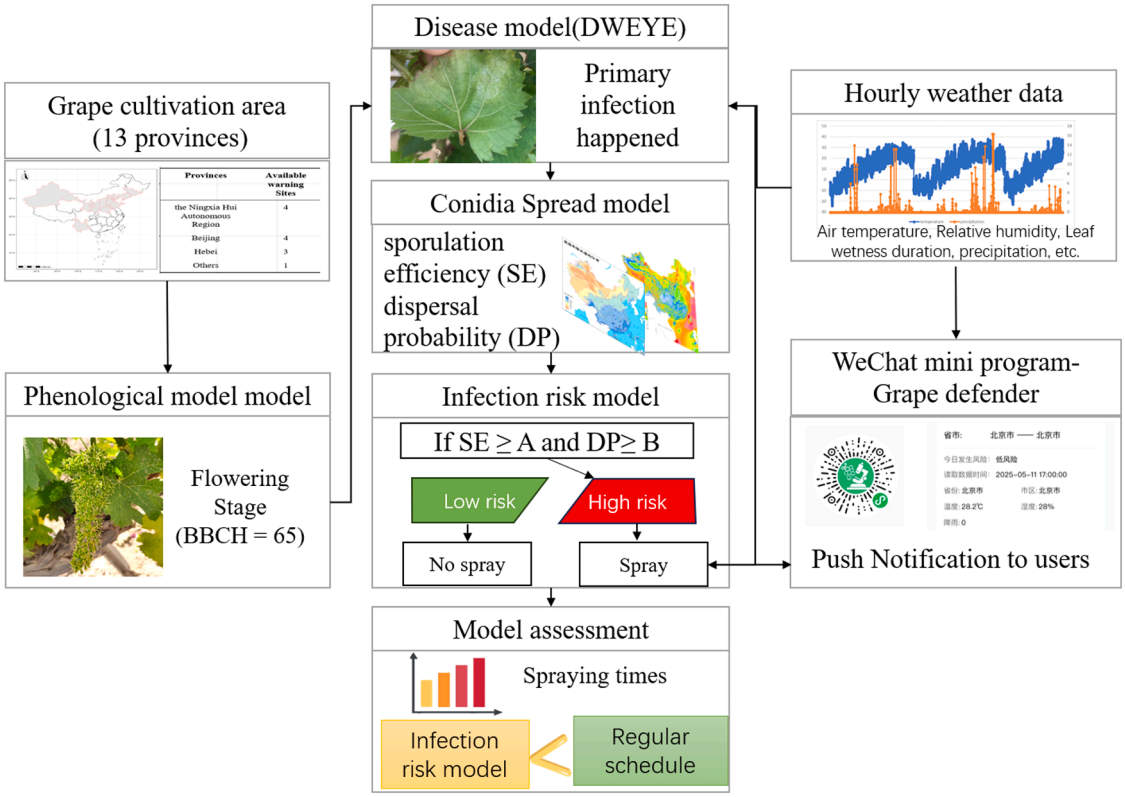


Fig. 1. Flowchart for grape downy mildew infection risk estimation and warning tool development.

institutes conduct surveys and analyses to provide disease and pest warnings. This information primarily addresses major diseases and pests affecting staple crops, typically encompassing wheat, rice, corn, potato, rapeseed (canola), legumes, and tomato. Moreover, in Europe, warnings are issued weekly, whereas in China they are released annually as forecasts of disease and pest trends rather than as precise early warnings.

The limited adoption of DSSs can be attributed to three main factors. First, governmental warnings are restricted in terms of target crops and temporal resolution. Second, some DSS interfaces are complex, preventing users from intuitively accessing clear information on whether pesticide application is required. Third, regional data limitations hinder large-scale validation and dissemination. To address these challenges, based on our previously developed forecasting model for grape downy mildew in China [16], we further developed it into a decision support system. Leveraging WeChat Mini Programs, a social software platform widely used in China, the system provides simple daily guidance on whether pesticide application is needed. This approach represents a successful case of a plant disease DSS delivering precise management services to 13 major grape-growing regions in China, illustrating the potential of locally adapted and user-friendly tools for improving disease management and promoting sustainable viticulture.

2. Materials and methods

The study flowchart is presented in Fig. 1. A total of 13 provinces were monitored to determine the grape downy mildew infection risk from 2021 to 2025 (Fig. 2). The flowering stage is a condition of pathogen infection in China [16]. We referenced a phenological model from Nogueira Júnior et al. [17] that estimates the grape flowering stage (GP, Eqs. (1), 2) using a piecewise equation comprising the daily average temperature (T_{day}) and base temperature (T_{base} = 10 °C), which determine when GP is over the flowering stage threshold (GP_t). GP_t is an average GP value simulated in China over multiple years (GP_t = 399 °C × day, [16]).

$$DD(T_{day}) = \begin{cases} T_{day} - T_{base} \ if \ T_{day} > T_{base} \\ 0 \ else \end{cases} \quad (1)$$

$$GP = \sum_{1}^{n} DD(T_{day}) \ \geq GP_t \quad (2)$$

Hourly weather data were collected from a meteorological website (https://airwise.hjhj-e.com/) and weather stations installed in grape-growing regions. These weather data were mutually calibrated to enhance data credibility and used to estimate the flowering stage and as inputs of the primary infection and conidia spread models.

A primary infection model developed in a previous study was used to estimate the date of disease occurrence [16], which is the first spraying time of the year for downy mildew control recommended to farms. The second step was calculating conidia spread, which depends on wind speed and precipitation. This work aimed to determine the need to spray during the subsequent growth period. When the values of wind speed and precipitation are over certain thresholds, the infection risk is high, and spraying should be performed. In contrast, when the infection risk is low, there is no need to spray. The thresholds were determined by comparing the number of sprays and incidence under multiple groups using MATLAB software (R2019b). The spraying frequency reflected the effectiveness of the infection risk model compared to a regular spraying schedule in several grapevine yards. Based on this, we developed a WeChat mini program named “Grape Defender.” Under high-risk conditions, the notifications automatically push users to spray. The 19 sites that were monitored to determine the grape downy mildew infection risk are listed in Table S1. Beijing, Hebei, and the Ningxia Autonomous Region included three monitored sites; in other provinces, only one site was monitored. From 2021 to 2024, Fangshan and Huailai were chosen to assess the effectiveness of the infection risk model compared to a traditional spraying schedule.

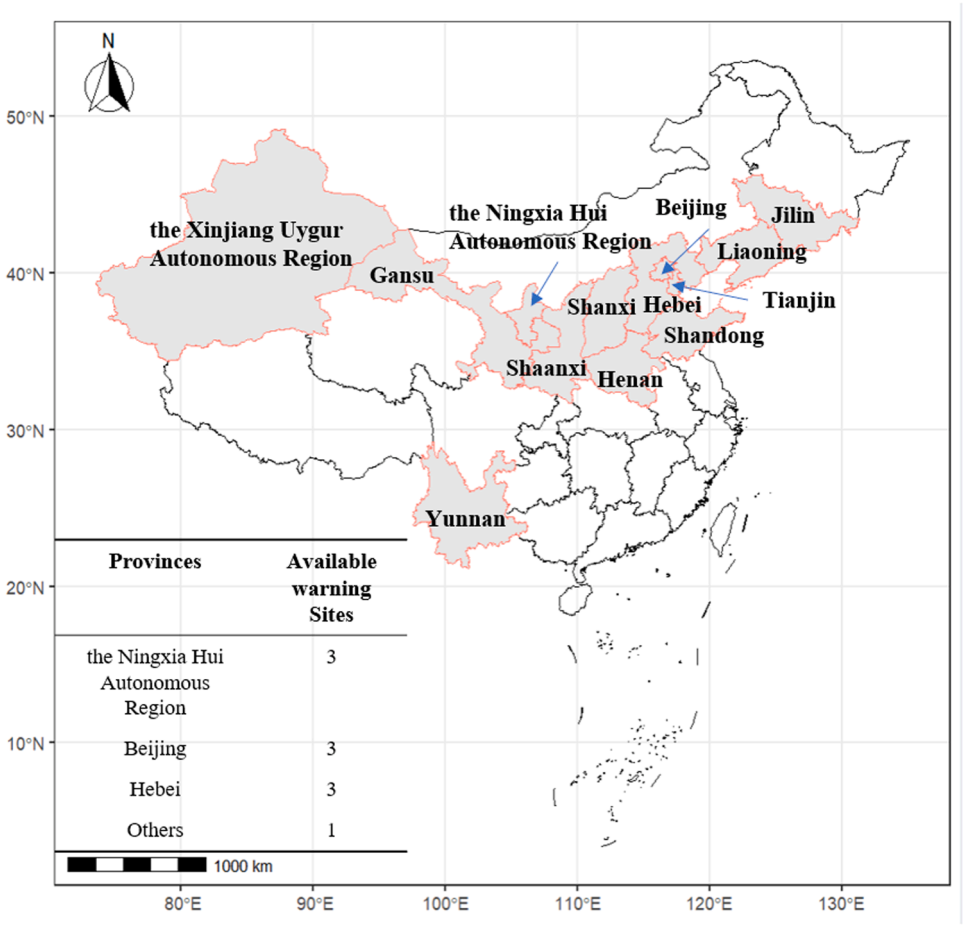


Fig. 2. Available warning regions of grape downy mildew in Grape defender WeChat Mini Program.

2.1. Conidia spread model

The conidia spread model considers sporulation efficiency (SE, unitless, ranging from 0 to 1, Eq. (3)–5) and dispersal probability (DP, unitless, ranging from 0 to 1, Eq. (4)) caused by precipitation. SE was computed as a function of the hourly air temperature (T_h) and relative humidity (RH). T_{max}, T_{min}, and T_{opt} represent the maximum, minimum and optimal air temperatures for sporulation, respectively ($T_{max} = 34$ °C, $T_{min} = 6$ °C and $T_{opt} = 15$ °C). RH_{min} is the minimum RH demand ($RH_{min} = 85.5\%$). The conditions for conidia spread were determined in our previous work [16].

$$A = \frac{T_{max} - T_h}{T_{max} - T_{opt}} \tag{3}$$

$$B = \frac{T_{opt} - T_{min}}{T_{max} - T_{opt}} \tag{4}$$

$$SE = \begin{cases} A \times \left(\frac{T_h - T_{min}}{T_{opt} - T_{min}}\right)^B \ if\ T_h \geq T_{min}\ and\ T_h \leq T_{max} and\ RH \leq RH_{min} \\ 0\ else \end{cases} \tag{5}$$

The extent of splashing rain (Rain, mm) determined DP [18], which was computed using Eq. (6), considering the leaf area index (LAI, m^2m^{-2}) and the precipitation required to detach 50% of the spores at a given LAI ($Pre_{50} = 1.62$, mm). The degree-day (DD) equation with a daily average temperature condition boundary ($T_{daily} \geq 10$ °C) from Williams and Ayars [19] was used to calculate the LAI (Eq. (7)).

$$DP = \frac{Pre}{Pre_{50} + LAI + Pre} \tag{6}$$

$$LAI = -2.6 + 6.63 \times \left(1 - e^{-0.0042 * DD}\right)\ T_{daily} \geq 10^{\circ}C \tag{7}$$

2.2. Infection risk model

The secondary infection cycle includes three stages: sporulation, dispersal, and infection. Among these, the sporulation and dispersal stages determine whether spores are produced and whether the produced spores can reach uninfected plant organs. Therefore, sporulation probability (DP) and spore dispersal efficiency (SE) were selected as indicators of secondary infection risk. Both variables range from 0 to 1, where values closer to 1 indicate a higher probability of sporulation and a greater likelihood of successful spore dispersal, thereby increasing the risk of secondary infection.

To determine appropriate risk thresholds, values of DP and SE within the range of 0.1–0.9 were evaluated using step sizes of 0.1 and 0.05 (Eq. 8). An initial threshold of 0.5 was used, where values ≥ 0.5 were considered high risk and triggered a spray warning, while values < 0.5 were considered low risk and did not generate warnings. Under different threshold combinations, the number of spray recommendations generated by the model was compared with the number of fungicide applications conducted under conventional vineyard management.

Preliminary results showed that when the threshold was set at 0.5, the number of spray recommendations generated by the model in 2022 and 2024 for wineries in Fangshan and Huailai exceeded the number of sprays applied under conventional management. Therefore, thresholds below 0.5 were excluded, and 0.5–0.9 was selected as the range for

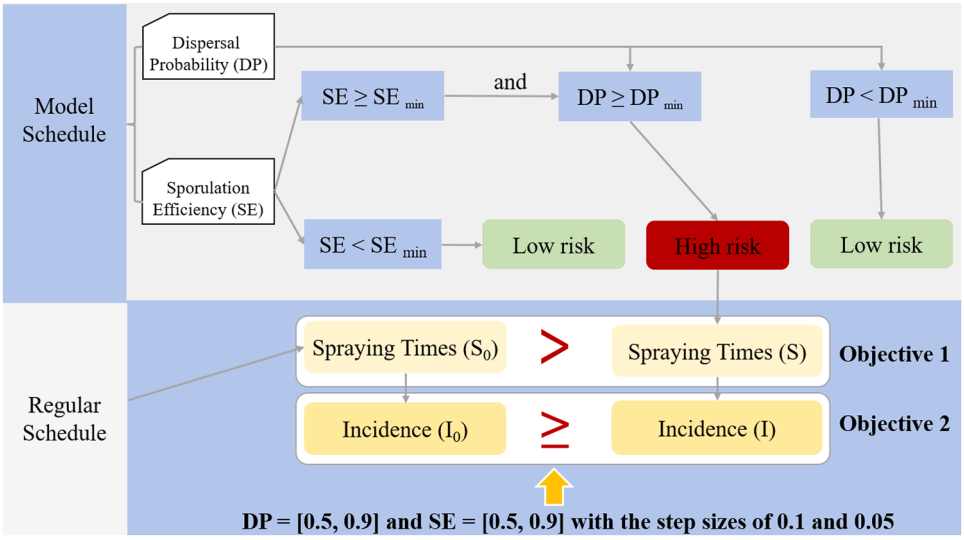


Fig. 3. The classification decision of infection risk of grape downy mildew.

further evaluation (Fig. 3). The grape downy mildew control method based on the model schedule should have a lower than or equal incidence but fewer sprays than the regular schedule (Eq. 9).

$$SE \geq SE_{min} \text{ and } DP \geq DP_{min} \quad 0.5 \geq SE_{min} \geq 0.9,\ 0.5 \geq DP_{min} \geq 0.9 \tag{8}$$

$$S_0 > S \text{ and } I_0 \geq I \tag{9}$$

2.3. Model assessment

Spray frequency was used as the indicator for assessing the infection

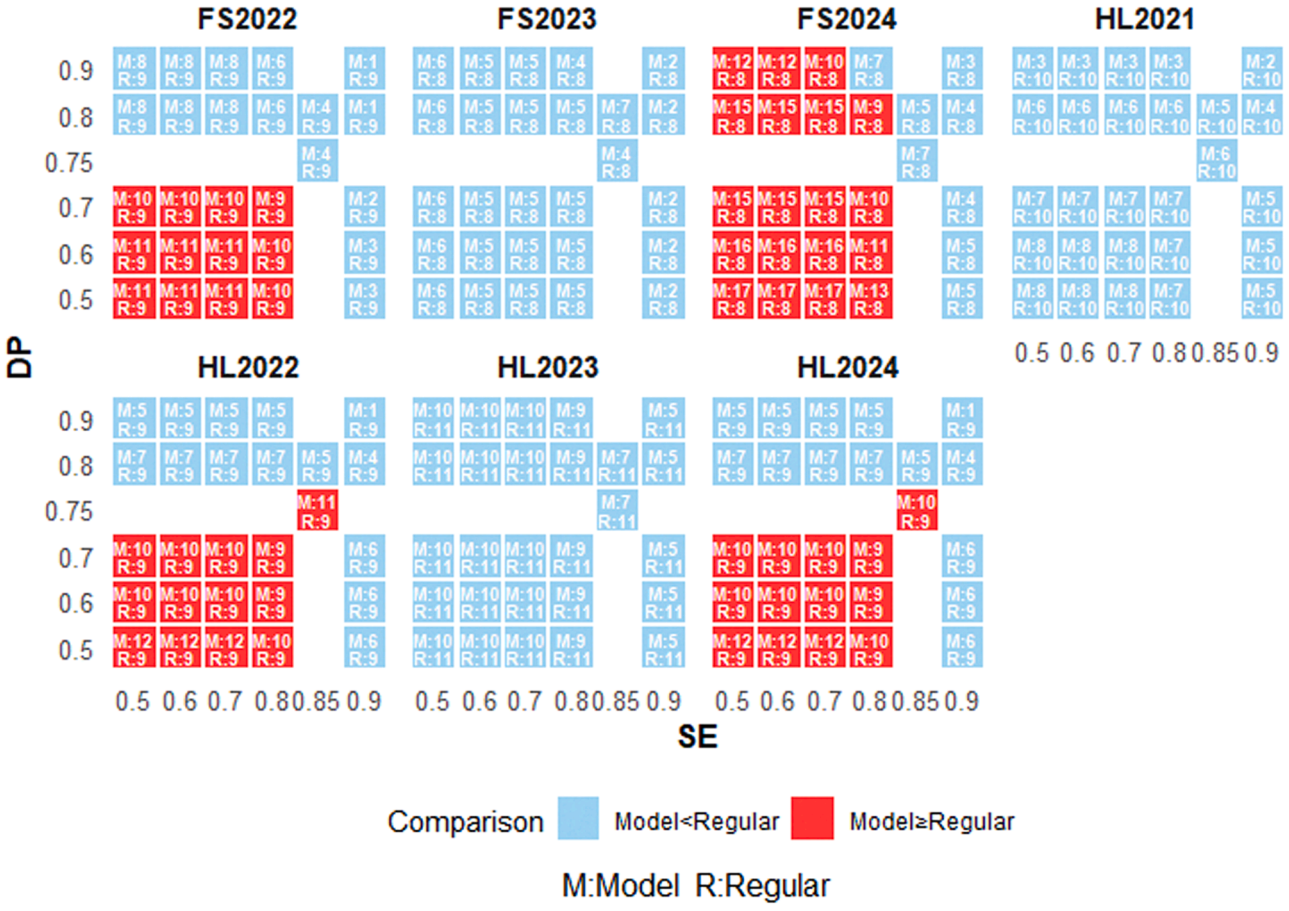


Fig. 4. Comparison of the number of sprays based on the model and regular schedule at two sites from 2021 to 2024.The red boxes represent fewer sprays than those in the regular schedule, and the sky-blue boxes represent the opposite result.
Note: The term "FS2022″ is a combination of the acronym for the district Fangshan (FS) and the numerical year (2022). "FS2023, FS2024, HL2021, HL2022, HL2023, and HL2024" follow the same format as "FS2022," where HL is the abbreviation for "Huailai".

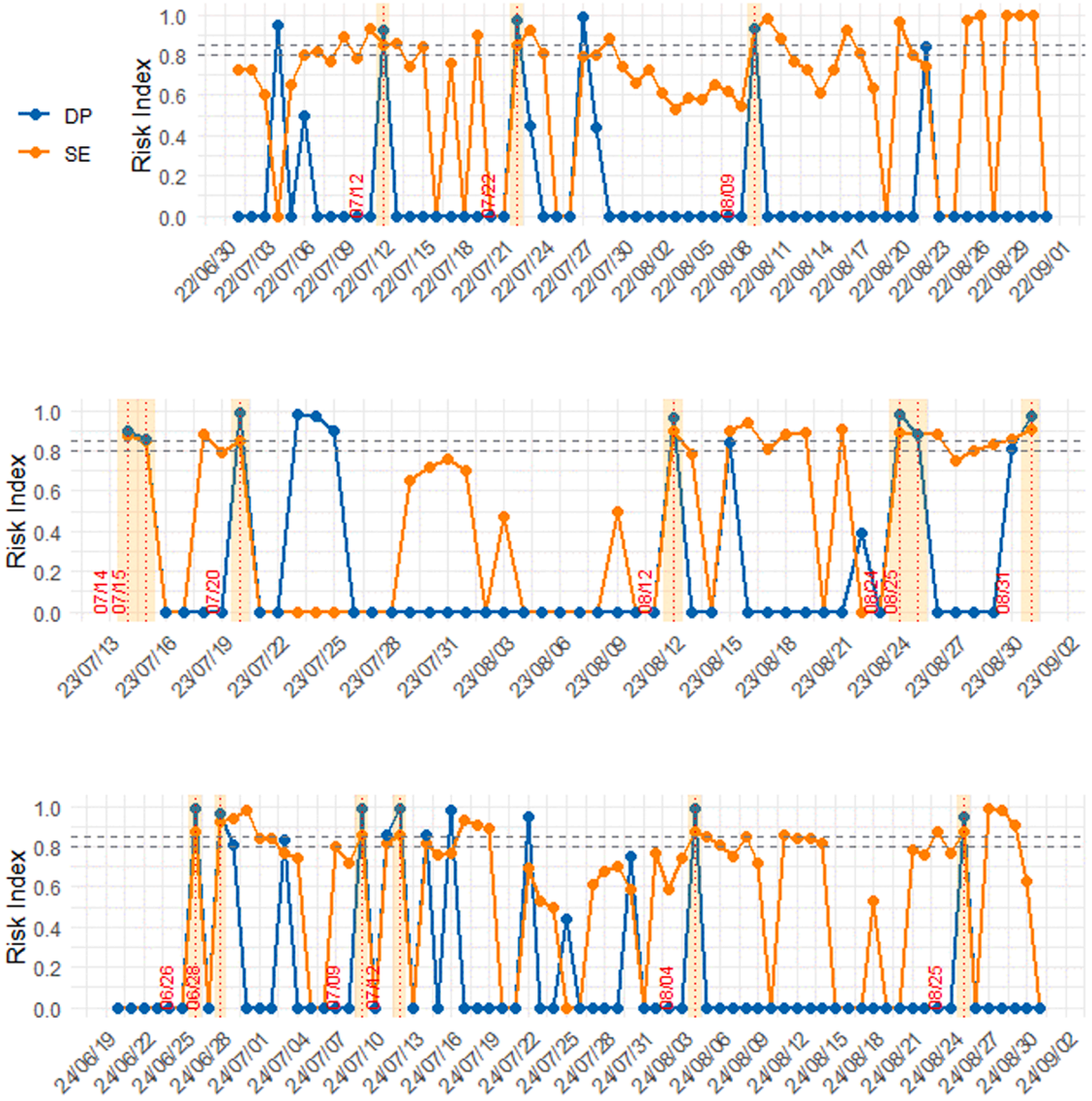


Fig. 5. Dynamics of daily dispersal probability (DP) and sporulation efficiency (SE) from June to August in Fangshan from 2022 to 2024 with high risk dates in red font (SE > 0.85 and DP > 0.8).

risk model via comparison with a traditional spray schedule at two sites from 2012 to 2024. The two sites were Fangshan (116.0078609°N, 39.70419597°E) and Huailai (40.413261°N, 115.5113854°E), representing grapes cultivated in the Pohai Coastal Region. One plot was chosen at each site. Half of the plot was an experimental trial, in which spraying was carried out depending on the risk model output, and the other half used the regular grape downy mildew control method. The plot area was 0.33 ha.

2.4. WeChat mini-program "Grape defender"

WeChat Mini-Program "Grape Defender" adopted a front-end and back-end separation architecture. The front end was developed using the uni-app framework combined with the uView component library to ensure cross-platform compatibility. The back end was built with Java based on the Spring Boot framework, integrating Spring Security and JWT for permission management and identity authentication.

For data storage, MySQL 8.0 served as the primary database, with Redis employed to enhance caching performance. File storage was implemented using Alibaba Cloud OSS, while SMS functionality was realized through the Alibaba Cloud SMS service. Message push notifications were configured based on the WeChat Mini Program Template Message mechanism. The development environment requires tools, such as a Java IDE (e.g., IntelliJ IDEA), Node.js, and WeChat Developer Tools, and the correct installation of corresponding dependency libraries for both the front end and back end.

The input port was the hourly air temperature, relative humidity, and precipitation at each site. The output port was the result of the infection risk model. Finally, the users can see the notification online or subscribe to notifications when the result is high risk. They have to spray on the following day when they have been notified of high risk. The program supplies climate data and the results of the infection risk model for three months.

3. Results

3.1. Analysis of grape downy mildew infection risk

The disease model (DWEYE) developed in a previous study estimated the occurrence dates for grape downy mildew. In Fangshan, these dates were July 1st, July 13th, and June 19th over the three years (2022–2024). In Huailai, the grape downy mildew occurrence dates

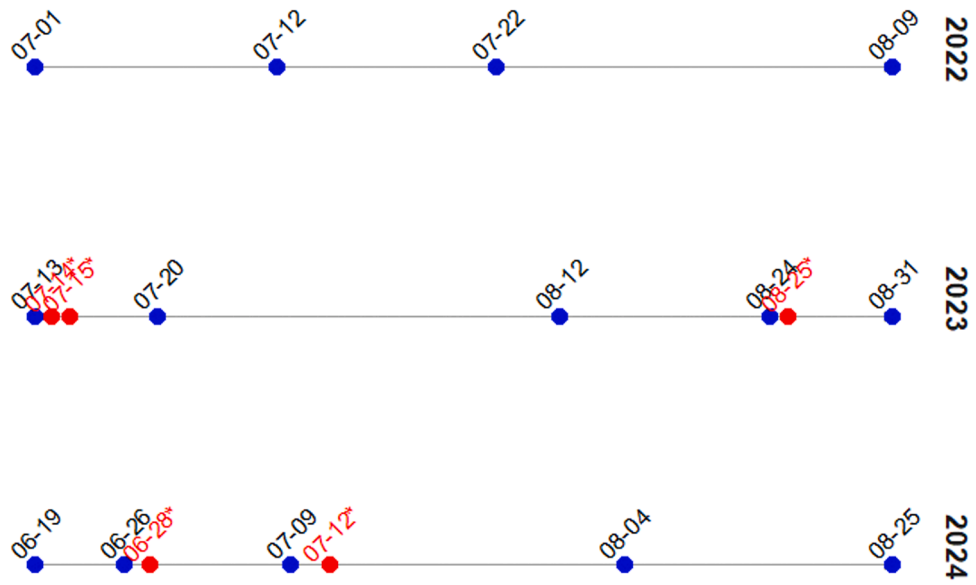


Fig. 6. Spraying dates in Fangshan from 2022 to 2024 depending on the high risk dates obtained by the model. No sprays were applied on the dates with red font because it was <7 days from the presious warning.

were June 30th, July 3rd, June 19th, and July 16th over the four years (2021–2024).

After disease onset, the risk model evaluated the daily infection risk caused by rainfall and sporulation. The thresholds of daily DP (0–1) and SE (0–1) were chosen from 0.5 to 0.9 with a step size of 0.1. When their values were higher than the thresholds, the day was considered to have a high risk for pathogen infection. We compared the number of sprays between the infection risk model and regular schedule in Fangshan and Huailai from 2021 to 2024 (Fig. 4). When DP was over 0.8 and SE was over 0.9, all groups showed fewer sprays based on the infection risk model than those in the regular schedule. Subsequently, when we reduced the step size of the SE and DP thresholds to 0.5, the model-based recommended fewer sprays in all groups than those in the regular schedule at an SE of 0.85 and a DP of 0.8.

The daily DP and SE are shown in Figs. 5 and 7. Based on the thresholds of DP ($DP_{min} = 0.8$) and SE ($SE_{min} = 0.85$), there were 3, 7, and 6 days of high risk for conidial secondary dispersal after grape downy mildew occurrence in Fangshan from 2022 to 2024 (dates in red font in Fig. 5) and 5, 8, 12, and 8 days in Huailai from 2021 to 2024 (dates in red font in Fig. 7).

Considering the safety regulations for pesticide application, although the model evaluated continuous high-risk outcomes, when the interval between dates was <7 days, it was recommended that the date of the previous warning be adopted as the spraying time. Consequently, there will be fewer spraying days than high-risk days. For example, at Fangshan, no spraying was required on July 14th or 15th or August 25th in 2023 or on June 28th or July 12th in 2024 (Fig. 6). The first spraying occurred after the warning for grape downy mildew, and the number of sprays was 4, 5, and 5 at Fangshan from 2022 to 2024 and 5, 5, 7, and 5 at Huailai from 2021 to 2024 (Figs. 6 and 8).

3.2. Comparison of spraying times between two grape management methods

In Fangshan and Huailai, the managers of the vineyards start to control for grape downy mildew in the middle to end of May. Then, a regular control schedule was carried out from June to August depending on the weather (Table S2). When it is rainy, they spray every 7–10 days. Spraying was typically carried out once in May at each site, but it was carried out twice at Fangshan in May 2023, and spraying was not performed at Huailai in 2024. There were no notifications of high risk in May; thus, spraying was not performed in the experimental plots (Fig. 9).

In June, 0–2 sprays were carried out at the two sites each year based on the outputs of the risk model, which was 1–4 fewer sprays than those used in the regular control method. In July and August, the managers pay more attention to grape downy mildew and spray more frequently than in May because the rainy season increases the disease incidence. However, 1 or 2 fewer sprays were applied using the risk model schedule, which occurred in August 2021 and July 2022 in Huailai.

The number of sprays was reduced to 5, 6, and 1 spray from 2022 to 2024 at Fangshan, compared to 12 sprays. At Huailai, 4 or 5 fewer sprays were applied each year than those used in the regular schedule, which applied a total number of 17 sprays. The disease incidence of the two control methods had similar results in most years, showing ranges of 3.1–20% using the regular method and 4.5–19% using the risk model method at Huailai and 8.9–100% and 7.8–71%, respectively, at Fangshan. This results demonstrate that reducing the use of pesticides in an appropriate manner did not increase the incidence.

3.3. Simulated results of grape downy mildew infection risk

The program monitors the onset dates of grape downy mildew and the high-risk dates for secondary infection in 13 provinces in 2025. Beijing, Hebei, and the Ningxia Autonomous Region included three sites, and only one site was monitored in the other provinces. Fig. 10 shows the first onset dates of grape downy mildew at all sites in each province. Grape downy mildew did not occur in the Xinjiang Uygur Autonomous Region in 2025. Yunnan was the only site with grape downy mildew in early June and had the most days with high-risk notifications. The program warned of the occurrence of grape downy mildew in mid-June in Beijing, Shandong, and Shanxi. After mid-June, Shanxi had more high-risk notifications in July and August. There were more high-risk notifications in late June in Shandong. Beijing only had one high-risk day in June and August. Late June was the period of grape downy mildew onset in Liaoning, Tianjin, Henan, and Hebei. In July and August, Hebei had three high-risk days in each month, and Liaoning had three high-risk notifications in two months. Tianjin and Henan had one and two days with high-risk notifications in June. For Shaanxi, Jilin, and the Ningxia Autonomous Region, grape downy mildew was predicted in early July in the first two cities and in late July in the third city. Although the date of grape downy mildew occurrence

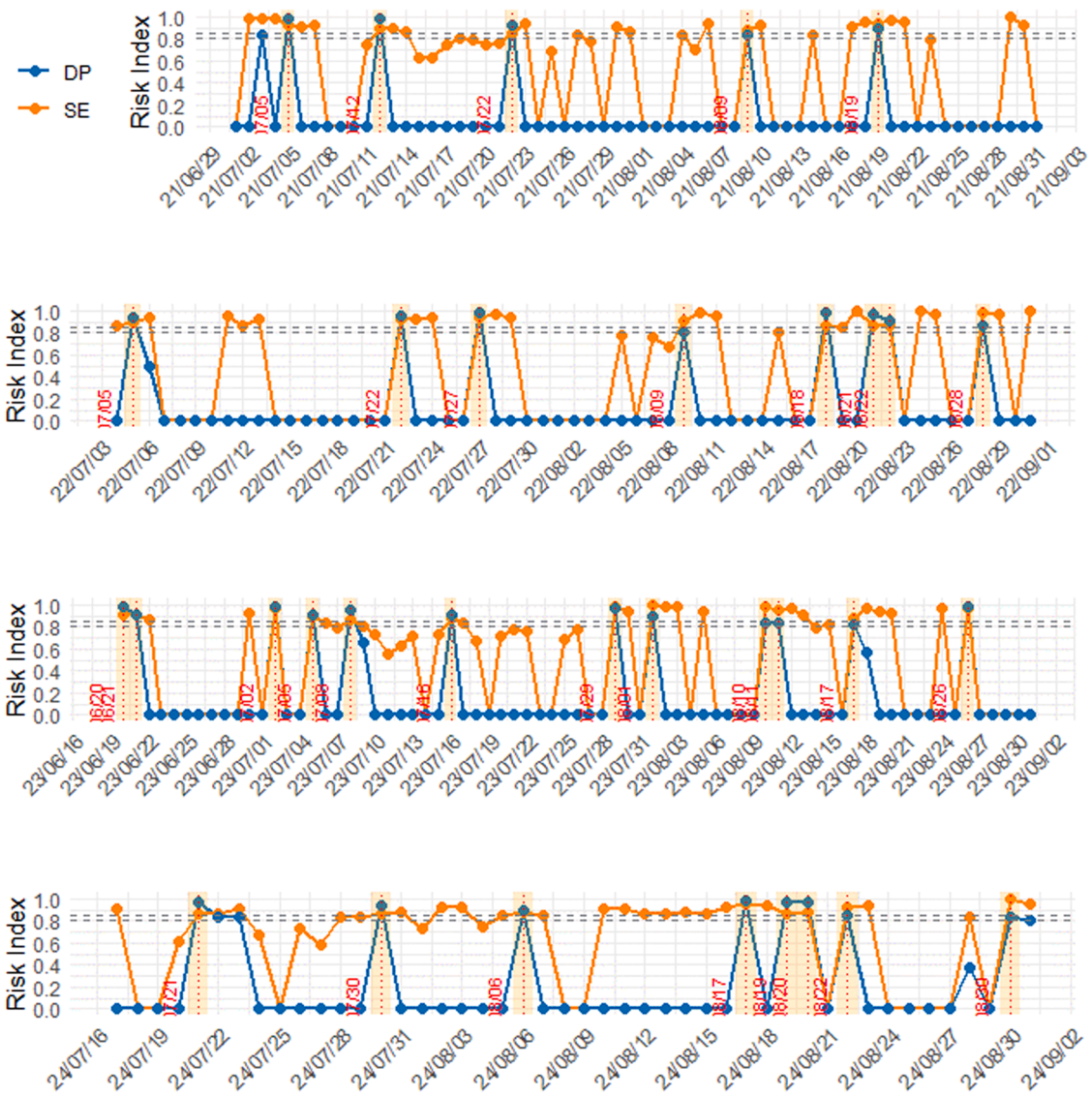


Fig. 7. Dynamics of daily dispersal probability (DP) and sporulation efficiency (SE) from June to August in Huailai from 2021 to 2024 with high risk dates in red font (SE > 0.85 and DP > 0.8).

was late in Jilin, there were more high-risk notifications in the later stages. July was a high-risk period for pathogen infection in Ningxia. Shaanxi only had one high-risk day in July. Compared to all other provinces, Gansu received the latest downy mildew warning in early August. Farms only need to pay attention to grape downy mildew in August. There was three high-risk days in August.

3.4. Introduction for using grape defender mini-WeChat program

We developed the "Grape Defender" mini-program to display the risk levels for grape downy mildew infection, the date of first occurrence, and relevant weather conditions in major grape cultivation regions and specific vineyards equipped with micro-weather stations across China (Fig. 11). To find and use the program, select the "Discover" tab at the bottom of WeChat, enter the "Mini Programs" section, and type "Grape Defender" in the search bar.

The specific instructions for use are as follows:

1. Register and log in to your account.
2. Click on the "Disease Warning" window.
3. Select the desired city by clicking on its location within the map.
4. View the current downy mildew infection risk level and environmental conditions. This shows the estimated incidence rate of downy mildew in major grape-growing regions of China assuming that no disease control measures are implemented.
5. Click on the name of the weather station you have installed and link it to the mini-program.
6. Browse the analyzed infection risk level for one week, one month, or three months based on data from the weather station and the estimated date of downy mildew onset in the current year.
7. Enable the subscription feature to receive ongoing updates.

4. Discussion

4.1. Scoring scheme for grape downy mildew infection risk

China Meteorological Administration published a meteorological industry standard called "Method for Forecasting Crop Disease Meteorological Grades." This is a 4-point item used to assess the weather risk based on plant disease model outputs and is classified as very high, high, relatively high, and low (China Meteorological [20]). A 5-point Likert response item was used to assess grape downy mildew infection risk in

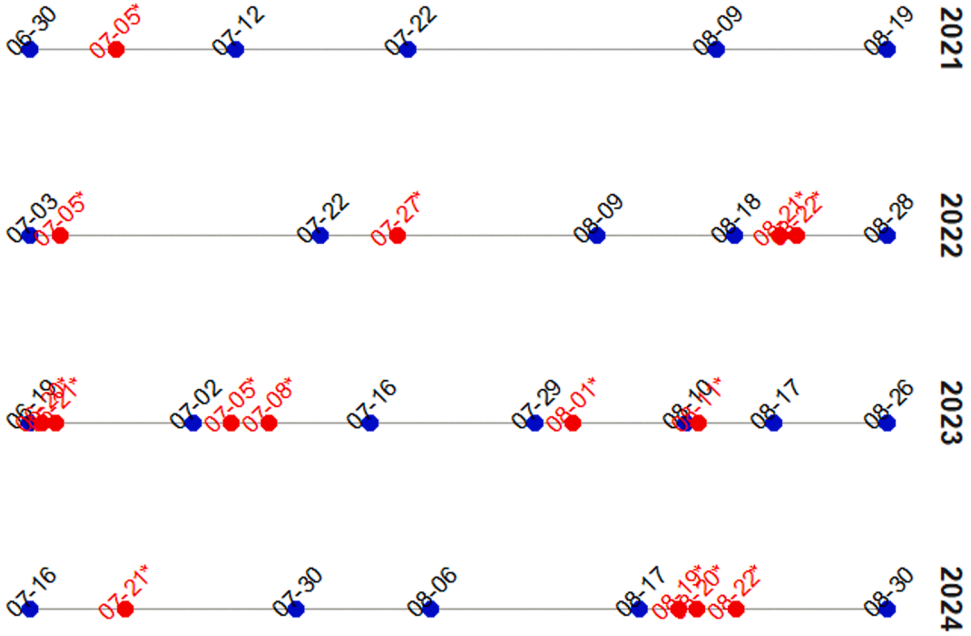


Fig. 8. Spraying dates in Huailai from 2021 to 2024 depending on the high risk dates obtained by the model. No sprays were applied on the dates with red font because it was <7 days from the presious warning.

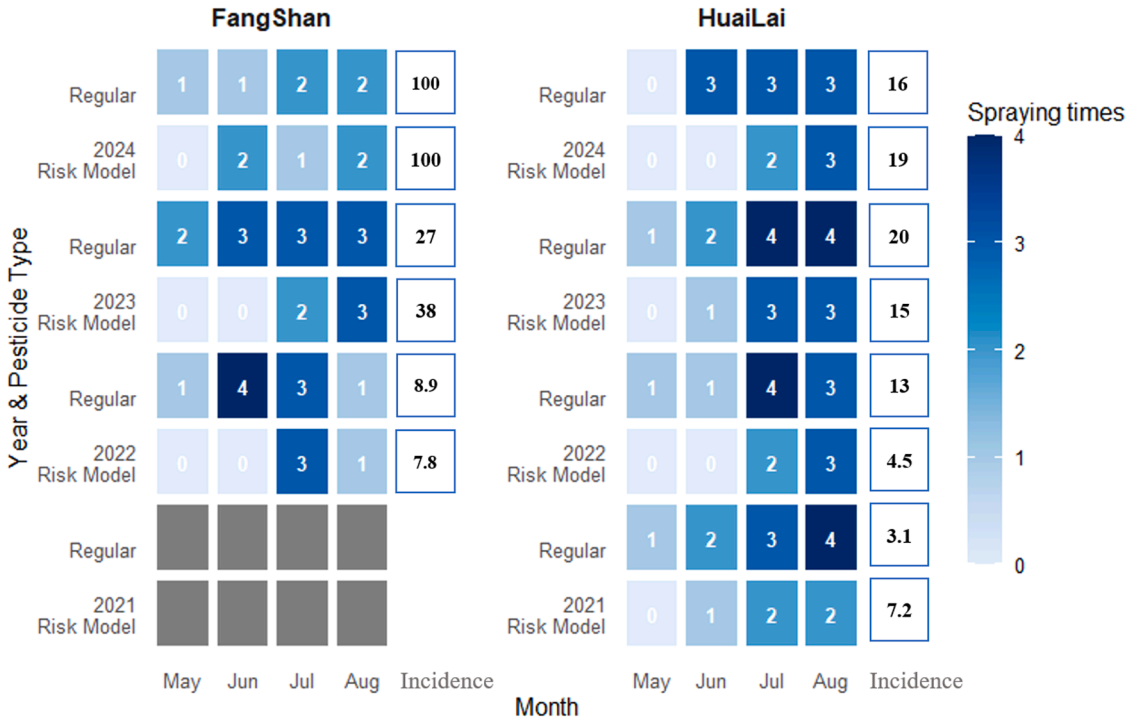


Fig. 9. Comparison of the number of sprays and the incidences of grape downy mildew recommended by the infection risk model with regular management of vineyards at Fangshan and Huailai over several years.

Italy based on people's opinions across disciplines, including psychometrics [21] and agro-environmental management [22]. In this study, a much cleaner risk level standard (2-point item: high and low risks) was proposed to serve as the basis for growers to make decisions on whether to apply pesticides. The multilevel risk level standard actually hinders growers' judgment.

4.2. *Thresholds of the infection risk model*

It is important to have an accurate threshold as the demarcation point for dividing infection risk levels. The daily DP and SE are values in the range of 0–1 and are used to assess the grape downy mildew infection risk. We divided the high- and low-risk thresholds equally. Thus, the threshold of high-risk days was between 0.5 and 0.9. By comparing the number of sprays based on high-risk days to the regular schedule, we obtained accurate thresholds for DP and SE of 0.8 and 0.85, respectively.

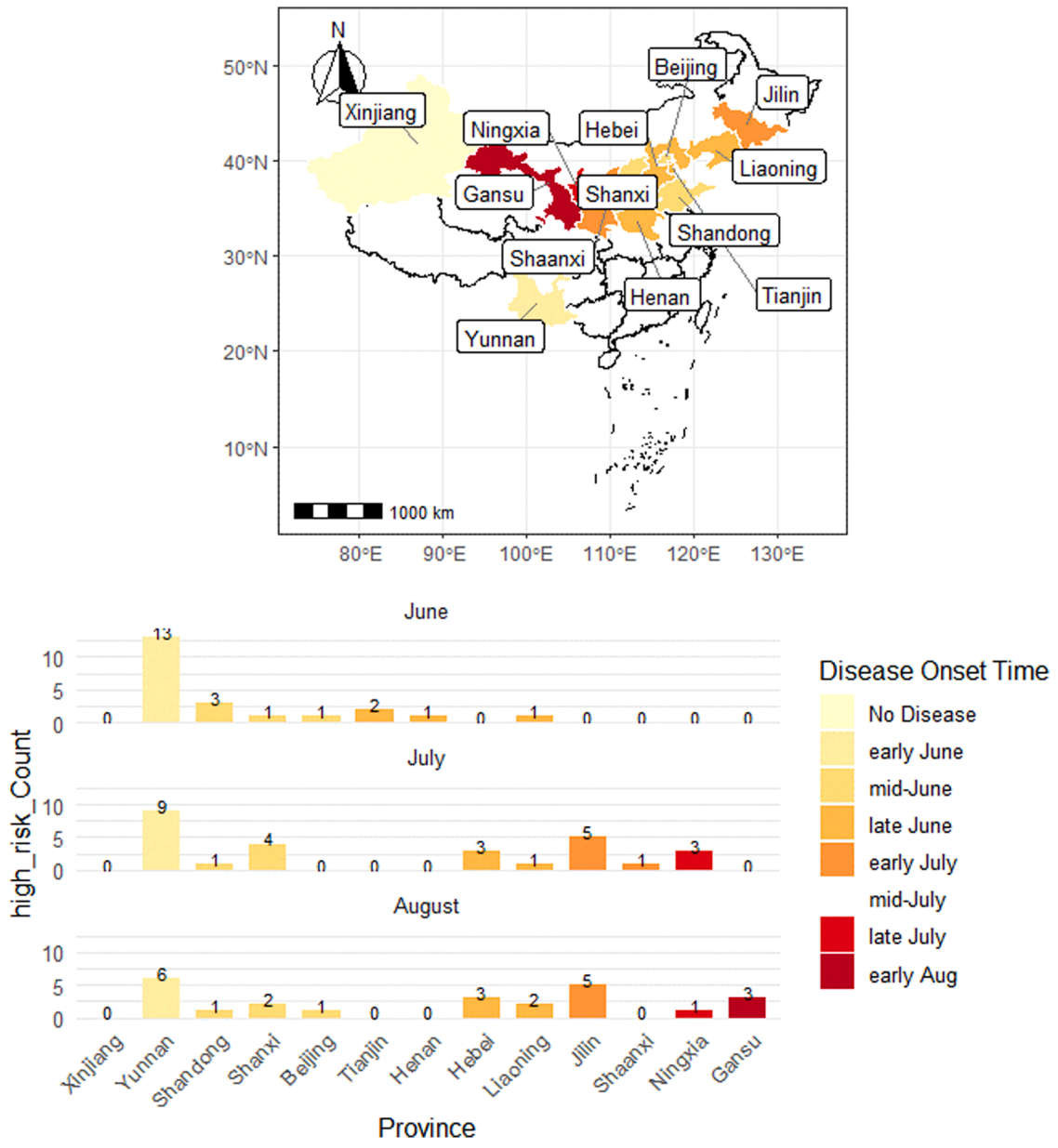


Fig. 10. Period of grape downy mildew onset in the monitored regions in 2025 determined by the "Grape Defender" Mini-WeChat program and the number of high-risk warnings by the infection risk model.
Notes: Ningxia is the short name for the Ningxia Autonomous Region; Xinjiang is the short name for the Xinjiang Uygur Autonomous Region.

Some traditional methods have been mentioned in a previous study [14]. For example, three increased incidence rate levels (1–2, 3–4, and 5–6%) were used in the field to test fungicide application timings on pod rot incidence compared to the regular calendar method [23]. Adding a control group, four plots were needed in each year to obtain the test results. We determined the optimal threshold via statistical analysis and then conducted field experiments for verification. This reduced the time cost and enabled more effort toward expanding the verification sites.

4.3. Reduction in the number of sprays without any noticeable impact on the disease incidence

In 2015, the Chinese government proposed an action plan for reducing chemical pesticides by 2025, aiming to reduce the need for pesticides by 5–10% compared to 13th Five-Year Plan period (2015–2020). Previous research has focused on developing DSSs to help farmers schedule fungicide sprays [24] and to determine the influence of climate change on plant disease risk [25]. Lázaro et al. [14] summarized the use of DSSs, showing a meaningful 43% reduction in sprays without any noticeable impact on the disease incidence across 22 studies. In this study, we reduced fungicide application by 43.75% across a 4-year period at two experimental sites based on the infection risk model. This research has demonstrated the reliability of the DSSs. Compared to the control group, the difference in disease incidence was <1.1–11%, except at Fangshan in 2024.

4.4. Current limitations and future directions for the "Grape defender" DSS

In terms of user services, the Grape Defender decision support system currently covers the major grape-growing regions in 13 provinces of China. For vineyards without on-site weather stations, the system provides risk warnings for grape downy mildew based on temperature and humidity data from municipal meteorological stations, which are

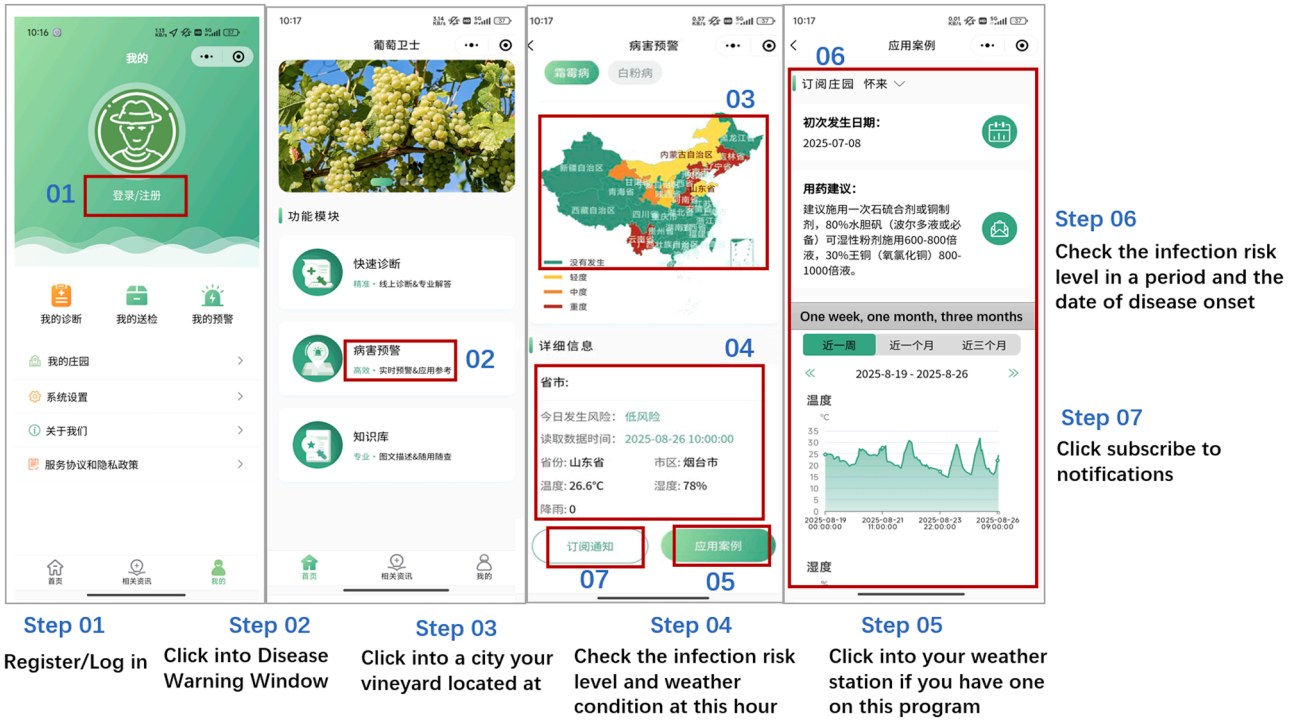


Fig. 11. The flowchart for the grape defender mini-program usage.

displayed on a map for reference by wineries and grape growers. For users who require higher spatial resolution of meteorological data, the system also allows local weather stations to be connected to the backend, enabling more precise warnings based on field-level weather conditions. However, these functions are currently more suitable for large-scale vineyards or wineries equipped with meteorological devices, while their applicability for smallholder farmers remains limited, which may restrict the large-scale adoption of the decision support system.

During the 2024 field demonstration in Fangshan, the incidence of grape downy mildew reached 100% in both the control and demonstration plots. Further analysis suggested that this outcome was likely related to inappropriate fungicide selection, which failed to effectively control disease development. This result indicates that, in practical applications, decision support systems should not only focus on the accuracy of model-based disease warnings, but also consider the compatibility between fungicide selection and the stage of disease development. Therefore, future research should systematically evaluate the effectiveness of different fungicides at various disease development stages through field experiments and incorporate these results into the system to provide more targeted recommendations for fungicide use.

Moreover, the current system still has room for improvement in terms of user interface design and information presentation, which could further enhance usability and acceptance among growers. Future work will expand field validation across more regions, continuously optimize model parameters, and improve system functionality and interface simplicity to enhance its applicability and promote adoption in different grape-growing regions.

5. Conclusions

This research supplies a helpful DSS to assess the grape downy mildew infection risk, reducing the number of sprays by 43.75% and not increasing the disease incidence by >11%. The DSS was developed into a WeChat program named “Grape Defender.” Growers can receive notifications when grape downy mildew occurs for the first time during the season and on high-risk days in the secondary infection cycle period. This fills the gap for a domestic decision-support system for grape powdery mildew and implements the fungicide reduction policy of fungicides in China. The results highlight June to August as critical months for grape downy mildew control, rather than May, suggesting a shift in seasonal management priorities.

Ethics statement

Not applicable: This manuscript does not include human or animal research.

CRediT authorship contribution statement

Hui Wang: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Wei Zhang:** Writing – review & editing, Funding acquisition, Data curation. **Mei Liu:** Investigation, Data curation. **Senli Jia:** Investigation, Data curation. **Junbo Peng:** Investigation, Data curation. **Caiping Huang:** Investigation, Data curation. **Jiye Yan:** Writing – review & editing, Resources, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.atech.2026.102109.

Data availability

Data will be made available on request.

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